

Energy Homework 2

Name: _____

Date: _____

Directions: Show all work. For each example:

- Define the system and the reference point ($h = 0$).
 - Calculate the energy values at the listed instants.
 - Calculate the final velocities (if necessary).
 - Draw energy pie charts.
 - Decide whether mechanical energy and total energy are conserved.
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1 Conservation of Mechanical & Total Energy

Problem 1: Rolling cart down a ramp (no friction)

A 1.5 kg cart starts from rest at the top of a smooth ramp with a vertical drop of 2.4 m.

Define the system: _____

What is the reference point ($h = 0$): _____

Energies at instants: Top ($h = 2.4$ m), Midway ($h = 1.2$ m), Bottom ($h = 0$ m).

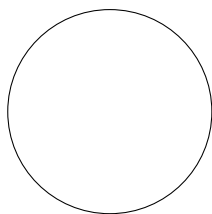
(a) Find the energy values at the top:

(b) Find the energy values midway down:

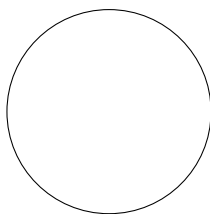
(c) Find the energy values at the bottom:

(d) Find the speed at the bottom:

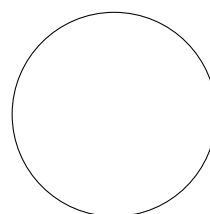
(e) Draw the energy pie charts:



Top of ramp



Midway



Bottom

(f) Is mechanical energy conserved?

(g) Is total energy conserved?

Problem 2: Pendulum swing (no air resistance)

A 0.80 kg pendulum bob is released from rest at a height of 0.60 m above its lowest point.

Define the system: _____

What is the reference point ($h = 0$): _____

Energies at instants: Release ($h = 0.60$ m), Mid-swing ($h = 0.30$ m), Lowest point ($h = 0$ m).

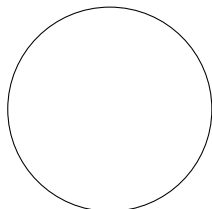
(a) Find the energy values at release:

(b) Find the energy values at mid-swing:

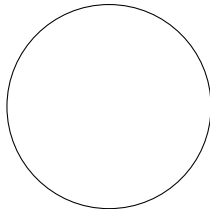
(c) Find the energy values at the lowest point:

(d) Find the speed at the lowest point:

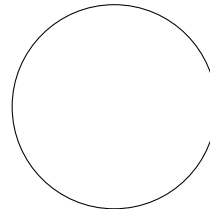
(e) Draw the energy pie charts:



Release



Mid-swing



Lowest point

(f) Is mechanical energy conserved?

(g) Is total energy conserved?

Problem 3: Spring launch on a level track (frictionless)

A cart of mass 0.70 kg is launched by a spring with $k = 160$ N/m. The spring is compressed $x = 0.18$ m and released on a frictionless level track.

Define the system: _____

What is the reference point ($h = 0$): _____

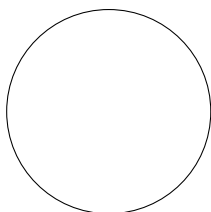
Energies at instants: At release (compressed), After release (spring relaxed, cart moving).

(a) Find the energy values at release:

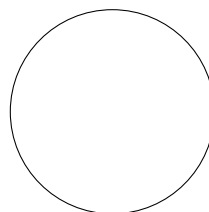
(b) Find the energy values after release:

(c) Find the cart's speed after it leaves the spring:

(d) Draw the energy pie charts:



Spring compressed



Cart moving

(e) Is mechanical energy conserved?

(f) Is total energy conserved?

2 Mechanical Energy Loss, Conservation of Total Energy Only

Problem 4: Dropped object (with air resistance)

A 1.6 kg ball is dropped from rest from a height of 12 m. Due to air resistance, it hits the ground with speed $v_{\text{impact}} = 13 \text{ m/s}$.

Define the system: _____

What is the reference point ($h = 0$): _____

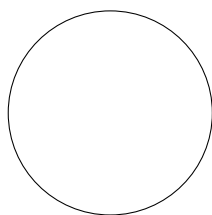
Energies at instants: Release ($h = 12 \text{ m}$), Impact ($h = 0 \text{ m}$).

(a) Find the energy values at release:

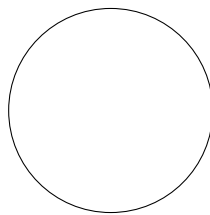
(b) Find the energy values at impact ($h = 0 \text{ m}$):

(c) Determine the thermal energy gained, E_{th} :

(d) Draw the energy pie charts:



Release



Impact

(e) Is mechanical energy conserved?

(f) Is total energy conserved?

Problem 5: Sled down a hill (with friction)

A 25 kg sled starts from rest at the top of a hill of height 4.0 m. Because of friction, the sled reaches the bottom with speed $v = 6.0$ m/s.

Define the system: _____

What is the reference point ($h = 0$): _____

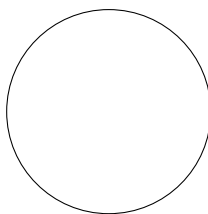
Energies at instants: Top ($h = 4.0$ m), Bottom ($h = 0$ m).

(a) Find the energy values at the top:

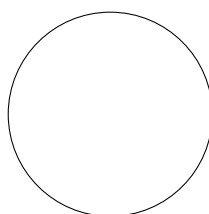
(b) Find the energy values at the bottom:

(c) Determine the thermal energy gained, E_{th} :

(d) Draw the energy pie charts:



Top of hill



Bottom of hill

(e) Is mechanical energy conserved?

(f) Is total energy conserved?

Problem 6: Spring launcher (with friction)

A cart of mass 0.60 kg is launched by a spring with $k = 180$ N/m. The spring is compressed $x = 0.22$ m and released. Because of friction, the cart leaves the spring region with speed $v = 2.8$ m/s.

Define the system: _____

What is the reference point ($h = 0$): _____

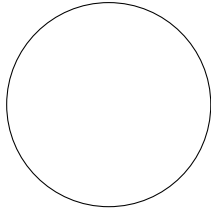
Energies at instants: At release (compressed), After release (spring relaxed, cart moving).

(a) Find the energy values at release:

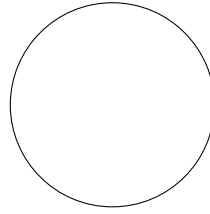
(b) Find the energy values after release:

(c) Determine the thermal energy gained, E_{th} :

(d) Draw the energy pie charts:



Spring compressed



Cart moving

(e) Is mechanical energy conserved?

(f) Is total energy conserved?